9–14. Graphing polynomials *Sketch a graph of the following polynomials. Identify local extrema, inflection points, and x- and y-intercepts when they exist.*

9. $f(x) = x^3 - 6x^2 + 9x$

10. $f(x) = 3x - x^3$

11. $f(x) = x^4 - 6x^2$

12. $f(x) = 2x^6 - 3x^4$

13. $f(x) = (x - 6)(x + 6)^2$

14. $f(x) = 27(x - 2)^2(x + 2)$

15–20. Graphing rational functions *Use the guidelines of this section to make a complete graph of f.*

15. $f(x) = \dfrac{x^2}{x - 2}$

16. $f(x) = \dfrac{x^2}{x^2 - 4}$

17. $f(x) = \dfrac{3x}{x^2 - 1}$

18. $f(x) = \dfrac{2x - 3}{2x - 8}$

19. $f(x) = \dfrac{x^2 + 12}{2x + 1}$

T 20. $f(x) = \dfrac{4x + 4}{x^2 + 3}$

T 21–32. More graphing *Make a complete graph of the following functions. If an interval is not specified, graph the function on its domain. Use a graphing utility to check your work.*

21. $f(x) = \sqrt{x}\,(x - 3)$

22. $f(x) = x^{1/3}(4 - x)$

23. $f(x) = x + 2\cos x$ on $[-2\pi, 2\pi]$

24. $f(x) = x - 3x^{2/3}$

25. $f(x) = x - 3x^{1/3}$

26. $f(x) = 2 - x^{2/3} + x^{4/3}$

27. $f(x) = \sin x - x$ on $[0, 2\pi]$

28. $f(x) = x\sqrt{x + 4}$

29. $f(x) = \dfrac{1 + x}{\sqrt{x}}$

30. $f(x) = x + \cos 2x$ on $[0, \pi]$

31. $f(x) = x + \tan x$ on $\left(-\dfrac{3\pi}{2}, \dfrac{3\pi}{2}\right)$

32. $g(t) = 3/t^2 - 54/t^4$